

A systematic review of Culturally Adapted Cognitive Behavioral Therapy (CA-CBT) for anxiety disorders in Southeast Asia

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ABSTRACT

Background: Cognitive Behavioral Therapy (CBT) has been proven an effective treatment for anxiety disorders. However, CBT still dominantly uses concepts and constructs rooted in Western cultures, and most research focuses on Western populations. It is unsure how this translates to non-Western cultures like Southeast Asia.

Aims: Our objective is to explore which types of cultural adaptations in CBT have been implemented for anxiety disorders in Southeast Asia and their effectiveness.

Methods: We systematically searched PubMed, PsycINFO, Embase, CENTRAL, GARUDA, and Google Scholar for CA-CBT for anxiety disorders in local communities in Southeast Asian countries. Data were analyzed using a narrative approach distinguishing between peripheral and core component adaptations. PROSPERO database preregistration number was CRD42022336376.

Results: Seven studies (one randomized controlled trial, three quasi-experimental studies, and three case reports) were selected. Two studies made cultural adaptations in multiple components. Two studies modified core treatment components by incorporating local values in the CBT restructuring process. Three studies conducted cultural adaptation on peripheral treatment components: adaptation to materials and semantics, cultural examples and themes, and session structure. Three studies did not provide detailed information. One RCT study showed better improvement for those who got CA-CBT than those in treatment as usual (TAU).

Conclusion: The findings suggest some components to consider when conducting cultural adaptation. We could not establish the degree of superiority of CA-CBT over non-CA-CBT nor identify components with the most influence due to the limited number of studies found. Employing standard documentation in reporting trials is also important to increase transparency.

1. Introduction

Anxiety disorders are among the most prevalent psychiatric disorders, with the highest illness burden and healthcare expenses (Bandelow and Michaelis, 2015). The World Health Organization (WHO) estimated that 60.05 million (23 % population prevalence) people live with anxiety disorders in Southeast Asia (World Health Organization, 2017). In Indonesia, around 8 million people (3.3 % of the population) suffer from anxiety disorders (World Health Organization, 2017). However, not all

people with mental illness in Southeast Asia, including Indonesia, get professional help due to the limited mental health services and human resources (ASEAN Secretariat, 2016).

The WHO Mental Health Gap Action Programme (mhGAP) has been initiated to scale up services to reduce the mental health gap (World Health Organization, 2008). One of the important factors in succeeding in scaling up the mental health intervention implementation is cultural adaptation (Faregh et al., 2019). Many critics highlight that the concepts and interventions developed in Europe or America are out of step with

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the diverse cultural values found in non-Western countries (White, 2013).

Cognitive Behavioral Therapy (CBT), one of the most studied psychological treatments worldwide and has been proven to have good outcomes across different delivery formats (Baker et al., 2021; Cuijpers et al., 2019), dominantly uses concepts and constructs rooted in Western cultures (Iwamasa and Hays, 2019; Naeem et al., 2016; Rathod et al., 2019). This condition may lead to the question of the efficacy of the treatment when delivered to the Asian population because of the different cultural values, beliefs, and attitudes (Rathod et al., 2018). Cultural differences may influence the process of psychosocial interventions (Hwang, 2006; Rathod and Kingdon, 2014).

To deal with the different cultural gaps, researchers proposed the idea of cultural adaptation (Cardemil, 2010; Castro et al., 2010). Cultural adaptation is defined as “the systematic modification of an evidence-based treatment (EBT) or intervention protocol to consider language, culture, and context in such a way that it is compatible with the client’s cultural patterns, meanings, and values” (Bernal et al., 2009, p. 362). One way to organize and map adapted components in the cultural adaptation of treatment is by using the classification of the Cultural Treatment Adaptation Framework (CTAF) (Chu and Leino, 2017). This framework divides adaptation components into two first-order categories: peripheral and core treatment components. Peripheral treatment components focus on making the treatment understandable and acceptable, while core treatment components focus on the main treatment components that mediate symptom change.

According to a review of meta-analyses conducted in 2018, research on culturally adapted interventions is still limited, lacks methodological rigor, and pays insufficient attention to theoretical foundations and cultural considerations. Moreover, most research explored the general therapeutic principles, which were not specific to CBT or any other specific intervention (Rathod et al., 2018). A previous finding also revealed that no clear definition and consistent evidence supports implementing any specific type of adaptation to its effectiveness (Healey et al., 2017).

Based on the explanation above, our current research aimed to systematically review empirical studies of Culturally Adapted CBT (CA-CBT), specifically in anxiety disorders focusing on the Southeast Asia region. A systematic review with this exclusive scope might help focus on revealing the face of cultural adaptation in Southeast Asia, which generally has socio-cultural similarities between countries (King, 2008). Understanding the cultural adaptation of CBT for anxiety in Southeast Asia is expected to gain in-depth information on why and how to develop a CBT protocol that will fit local people’s culture and context. In summary, this systematic review aimed to answer the following questions:

1. What types of cultural adaptations in CBT have been implemented in Southeast Asia for local communities?
2. What is the effectiveness of CA-CBT in treating anxiety disorders in local Southeast Asian populations?

2. Method

2.1. Approach and design of the review

This systematic review followed the guidelines outlined by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Moher et al., 2009) and was registered on the Prospective Register of Systematic Reviews (PROSPERO) (ID: CRD42022336376). For research question 1, this systematic review employed studies with various designs (including non-controlled pre-post studies) to thoroughly understand reported cultural adaptations across the literature. For research question 2, we only used randomized controlled trials (RCTs) that assessed the efficacy of CA-CBT.

2.2. Inclusion and exclusion criteria

Inclusion criteria considered (1) target population: Southeast Asian people with anxiety disorders under the classification of DSM-III-R, DSM-IV-TR, or DSM-5; (2) type of intervention: Culturally Adapted Cognitive Behavioral Therapy (CA-CBT); (3) study design: randomized controlled trials (RCTs) and non-randomized controlled trials (non-RCTs), including quasi-experimental studies, observational studies, and case reports; (4) the control group could receive treatment as usual, another active or passive treatment, placebo, waiting list, or no intervention; (5) outcome data was reported on anxiety severity; (6) published in English or Indonesian, in peer-reviewed journals with available full-text articles.

We referenced Southeast Asian countries according to the United Nations: Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand, Timor-Leste, and Viet Nam (United Nations, 2022; United Nations Statistics Division, 2022). Post-traumatic stress disorder (PTSD) was included since PTSD was regarded as an anxiety disorder before DSM-5. There were no restrictions on age, gender, or any other sociodemographic variables.

Exclusion criteria included the following: (1) studies using Southeast Asian people living outside their home country, for example, refugees; (2) CA-CBT combined with pharmacological treatments or other psychological treatments; (3) study protocol without data.

2.3. Search Strategies

This study followed Cochrane guidelines for conducting a systematic review (Higgins and Green, 2011). We searched in July 2022 and conducted an updated search in June 2023 in the following databases: PubMed, PsycINFO, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), Garba Rujukan Digital (GARUDA), and Google Scholar. Our search strategy included the combination of MeSH terms, Thesaurus, and free text words. No limitations were imposed regarding the papers’ publishing dates. We sought clarification from the corresponding authors in case of unclarity or missing information in the published study. The full search strategy is available in supplementary file 1.

2.4. Screening and data extraction

Titles and abstracts were screened by two authors (AP and AVI) independently. After completing the screening process, the full-text articles were collected and checked for their eligibility. Any discrepancies were resolved by discussing between two authors (AP and AVI) and involving the third and fourth authors (MHN and TKB) if needed. Extracted data included the authors, year, country, target population and sample size, study design, outcome measures used, intervention, type of cultural adaptation, delivery format, comparator, and outcomes.

2.5. Quality assessment

Since bias could affect the credibility of the evidence presented, we assessed the risk of bias in all studies included. Two reviewers independently assessed the risk of bias using the Cochrane risk-of-bias tool (RoB 2) for randomized trials (Sterne et al., 2019). For non-randomized studies, ROBINS-I was used (Sterne et al., 2016), and Joanna Briggs Institute’s (JBI’s) critical appraisal tool was used for studies with case report design (Moola et al., 2020). During the assessment process, disagreements between the review authors were resolved by discussion.

2.6. Data analysis

To address Research Question 1, a narrative synthesis was conducted following the Cultural Treatment Adaptation Framework (CTAF), which identifies two components, specifically peripheral and core treatment

components (Chu and Leino, 2017). For addressing Research Question 2, only RCTs were included. Due to the scarcity of trials, a narrative synthesis was undertaken because it was not feasible to do a meta-analysis. The results of the included studies were synthesized, summarized, and presented with no reference to the statistical significance of the findings (Siddaway et al., 2019).

3. Result

3.1. Search results

A total of 1653 records were retrieved from the search. After removing duplicates, 1073 records were included for titles and abstract screening. Then, we removed 1051 irrelevant records based on title and abstract, which led to 22 records that were identified as potentially relevant in the full-text review. Finally, seven studies (one RCT, three quasi-experimental studies, and three case reports) were found eligible for the review for data extraction. The study selection process is presented in Fig. 1 using the PRISMA flow diagram.

3.2. Study characteristics

All seven included studies were published between 2007 and 2022

(two published in 2007–2011; five published in 2017–2022). One study was a randomized controlled trial (RCT), three were quasi-experimental studies using pre-test and post-test designs, and three were case reports. Of the seven studies, three were conducted in Indonesia, two in Thailand, one in Singapore, and one in Malaysia. Furthermore, three studies focused on PTSD, two on social anxiety disorder, one on panic disorder, and one on multiple anxiety disorders. Three studies focusing on PTSD were included because they were published before the DSM-5 was released or referred to DSM IV/DSM-IV-TR, which still included PTSD as part of anxiety disorders. For detailed study characteristics, see Table 1.

3.3. Quality assessment

According to the reviewers' assessment, one randomized study was assessed in the category of some concerns. Three quasi-experimental studies were assessed to have a moderate risk of bias, and all three case studies were assessed as qualified enough to be included in this review. For detailed information, see supplementary file 2.

3.4. The design of the CBT

All CBTs were delivered face-to-face. Five studies were CBT delivered

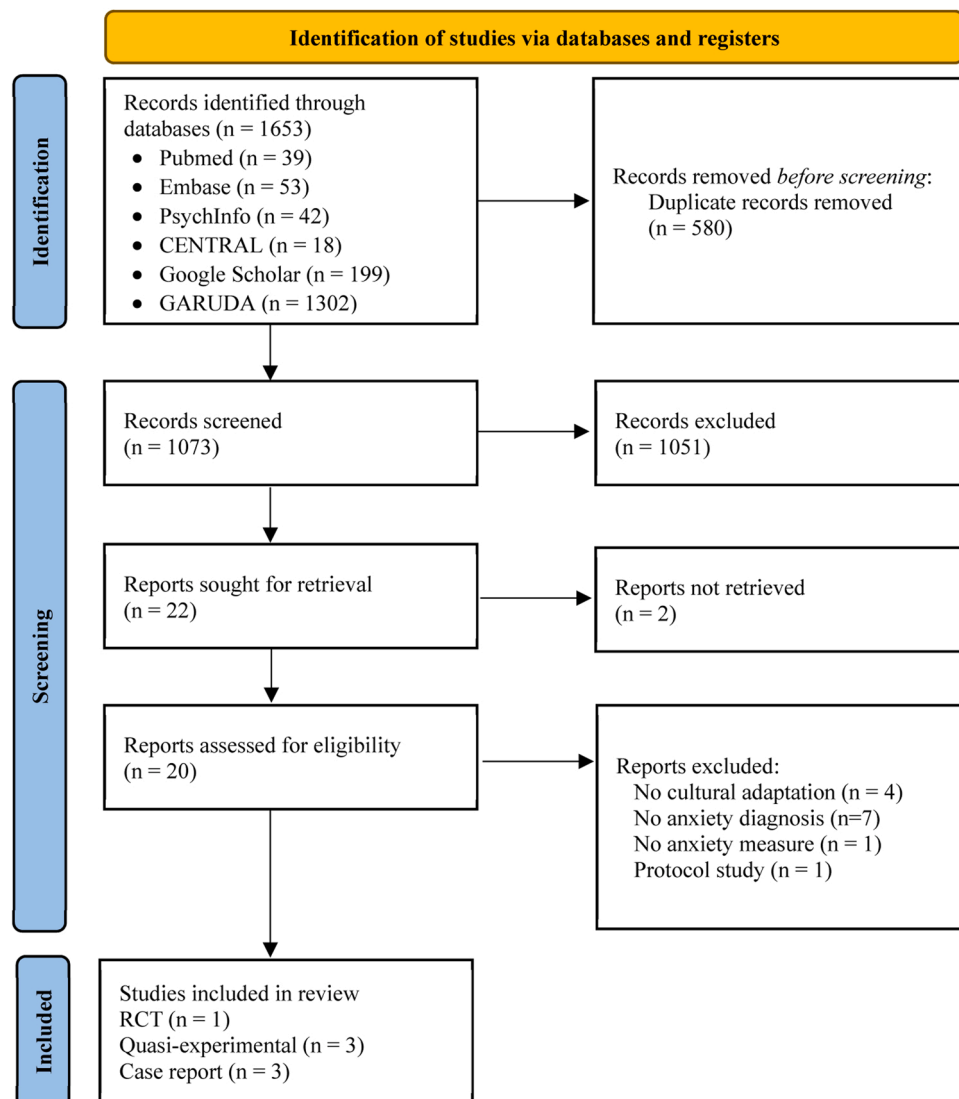


Fig. 1. PRISMA flow diagram of study selection.

Table 1
Study characteristics of included studies.

Authors	Country	Study design	Target Population	Outcome	Anxiety Measure	How to Diagnose Anxiety	DSM Diagnosis	Comparison	n	Summary of study findings
Bryant et al. (2011)	Thailand	Randomized controlled trial	Survivors of terrorist attacks	Symptoms of PTSD, Depression, Complicated Grief	The PTSD Symptom Scale - Interview (PSS-I)	PTSD diagnosis was determined by the PSS-I; this study used DSM-IV criteria in diagnosing PTSD.	Posttraumatic Stress Disorder (DSM-IV)	Treatment as usual (TAU) consisting of generic counseling	28 total (16 intervention; 12 control)	At the follow-up, CBT substantially outperformed TAU in terms of symptom reduction, including PTSD, depression, and complicated grief. At follow-up, CBT had larger effect sizes for PTSD, depression, and complicated grief compared to TAU. Compared to participants in the TAU (33 %), more patients in the CBT condition (75 %) had high end-state functioning.
Drmic et al. (2017)	Singapore	One group pre-test post-test design	Adolescents with autism spectrum disorder and anxiety	• Child-related anxiety disorder: panic/ somatic, generalized anxiety, separation anxiety, school avoidance • Developmental behavioral: disruptive/ antisocial, self-absorbed, communication disturbance, anxiety, social relating	• Screen for Child Related Anxiety Disorders (SCARED) – child and parent versions • Developmental Behavioral Checklist-Teacher (DBCT)	A screening questionnaire, the SCARED (child and parent) was used to select participants. Each student had to meet or exceed the cut-off for the Total Anxiety Disorder score or for at least one subscale score (i.e., Panic Disorder, Generalized Anxiety Disorder, Separation Anxiety Disorder, Social Anxiety Disorder, or Significant School Avoidance) on the child or parent version.	One of the diagnoses: Panic Disorder, Generalized Anxiety Disorder, Separation Anxiety Disorder, Social Anxiety Disorder (DSM 5)	No comparison	44 total	Significant decreases in anxiety symptoms were found both from the child's and parents' ratings after the intervention. However, there was no differences in the presence of emotional and behavioral issues using DBC-T.
Pityaratstian et al. (2007)	Thailand	One group pre-test post-test design	Children (aged 9-15 years)	Posttraumatic stress symptoms	Children's Impact of Events Scale (13) (CRIES-13)	The scores above the cut-off point (sum of Intrusion and Avoidance scores of 17 or more) suggest the significant impact of trauma on the child that the probability is very high that he/she will obtain the diagnosis of PTSD after formal diagnostic procedures; this study was published before DSM-V was released.	Posttraumatic Stress Disorder (DSM-IV-TR)	No comparison	152 total	No significant difference in CRIES scores was found between before and after the intervention. However, when the sample was categorized into two groups (CRIES+ve and CRIES-ve), there were significant differences. In the CRIES+ve group, significantly decreased was found in the Total, Intrusion, and Arousal scores. At the same time, in the CRIES-ve group, significantly increased was found in the Total,

(continued on next page)

Table 1 (continued)

Authors	Country	Study design	Target Population	Outcome	Anxiety Measure	How to Diagnose Anxiety	DSM Diagnosis	Comparison	n	Summary of study findings
Hayati et al. (2018)	Indonesia	Two-group pre-test post-test design	Adolescent (aged 12–18)	PTSD symptoms	Child PTSD Symptom Scale (CPSS)	Using the CPSS score to assess PTSD; in the article using the DSM-4-TR as the reference	Posttraumatic Stress Disorder (DSM-IV-TR)	CBT 5 sessions vs. CBT 12 sessions	38 total 19 group 1 19 group 2	Intrusion, and Avoidance Scores. No statistically significant correlation between practice days and mean difference in pre- and post-intervention CRIES scores was found. The PTSD score decreased in the CBT group of 5 sessions and the CBT group of 12 sessions. No significant difference in effectiveness was found between the two groups. CBT reduced anxiety symptoms substantially in the participant.
Subhas et al. (2021)	Malaysia	Single-case study	Malaysian Muslim with panic disorder with agoraphobia	- Anxiety symptoms - Panic attack symptoms	- Beck Anxiety Inventory (BAI) - Panic attack progress records	Diagnostic status was assessed using the BAI, mental state examination, clinical interview, physical examinations, and panic attack progress records.	Panic Disorder (DSM 5)	No comparison	1 total	
Wahyuningsih and Daengsari (2020)	Indonesia	Single subject A-B-A' design	Adolescents with Borderline Intellectual Functioning (BIF)	Social anxiety	Social Anxiety Scale for Adolescent (SAS-A)	Using observation, interview, and psychological examination to diagnose social anxiety. In addition, a score of SAS-A was used (the participant got a score of 85 out of 90)	Social Anxiety Disorder (DSM 5)	No comparison	1 total	The treatment decreased the social anxiety level of the participant.
Hayati and Tohari (2022)	Indonesia	A single-case experimental design	Students with social anxiety disorder and obsessive compulsive disorder	Social anxiety symptoms, obsessive-compulsive symptoms	Social Anxiety Scale (SAS)	Using the SAS category (high) in assessing social anxiety disorders	Social Anxiety Disorder (DSM 5)	No comparison	1 total	The participant showed significant differences in the frequency of social anxiety and obsessive-compulsive behavior before and after being given an intervention.

in an individual format, whereas the remaining used a group format. The number of CBT sessions varies, ranging from 2 to 15 sessions. The duration of the CBT ranged from 2 days to 15 weeks, with the length of each session ranging from 1 h to 2 h. For detailed information, see supplementary file 3.

3.5. Type of cultural adaptations

Of all studies, four explicitly reported cultural adaptation components, while three did not explicitly report the components of adaptations they made. Two studies considered multiple components of cultural adaptation (Drmic et al., 2017; Subhas et al., 2021). Of seven studies, two modified CBT's core treatment components (Bryant et al., 2011; Subhas et al., 2021), and three culturally adapted peripheral treatment components (Drmic et al., 2017; Hayati et al., 2018; Subhas et al., 2021). Three studies that provide undetailed information (Hayati and Tohari, 2022; Pityaratstian et al., 2007; Wahyuningsih and Daengsari, 2020) only stated performing adaptation but did not provide further information on which specific components or elements of CBT.

All cultural adaptations were made to improve its acceptability to the local communities. One example of cultural adaptation in the CBT technique as part of the core treatment components was integrating the Buddhist technique of distancing oneself from one's thoughts into cognitive restructuring (Bryant et al., 2011). Other examples focusing on peripheral treatment components were translating the language to Bahasa Malaysia (Subhas et al., 2021) and changing the example of fear in the graded exposure video (Drmic et al., 2017). Detailed information on identifying cultural components of the adaptation of the studies using CTAF can be seen in Table 2.

3.6. Cultural adaptation process

Of seven studies, only two provide information on how the adaptation process was carried out. For more detailed information, see Table 3. For the other five studies, we could not identify and synthesize the information from them due to the limited information provided.

3.7. Effectiveness of CA-CBT

This review was not feasible to continue to meta-analysis due to the scarcity of trials. This systematic review found only one RCT of CA-CBT in Southeast Asia (Bryant et al., 2011), which developed CA-CBT for survivors of terrorist attacks in Thailand. The participants who received culturally adapted CBT experienced a significantly higher reduction in PTSD scores than participants in the control condition who received treatment as usual (TAU) consisting of generic counseling.

4. Discussion

This study aimed to synthesize the literature on the type and effectiveness of CA-CBT. To the best of the author's knowledge, this is the first study focusing on the Southeast Asian population on CA-CBT for anxiety disorders. Based on our inclusion criteria, we found seven studies, including one RCT, three quasi-experimental, and three case reports. Of seven studies, only four explicitly reported the cultural adaptation components they made. Two studies modified CBT's core treatment components, and three conducted cultural adaptations on peripheral treatment components – two made cultural adaptations in multiple components. Due to the limited number of studies, this study could not establish the degree of superiority of CA-CBT over non-CA-CBT nor identify the components with the most influence.

We found interesting examples of cultural adaptation in the core component of the treatment, namely incorporating religious and cultural values in the CBT (Bryant et al., 2011; Subhas et al., 2021). Implementing cultural adaptation in peripheral components is reported in several types. First is modifying the materials and semantics by

Table 2

Type of cultural adaptation and examples.

Authors	Type of Adaptation	Cultural adaptation component and example
Bryant et al. (2011)	Core treatment components – modification	They modified relaxation with Thai meditation techniques and Buddhist meditation. They also integrated Buddhist techniques of distancing oneself from one's thoughts (in a Western, it might be called a mindfulness strategy) into cognitive restructuring to reduce a tendency toward catastrophic thinking.
Drmic et al. (2017)	Peripheral treatment components – materials and semantics Peripheral treatment components – cultural examples and themes	Translating the language and changing vocabulary to be more culturally appropriate Changing the example of fear in the graded exposure video from a video of a child facing his fear of dogs into a video of a child facing fear of 'talking on phone' and 'being in a store when it's closing' due to the advice from school administrators that there could be a cultural objection for some members of their community.
Pityaratstian et al. (2007)	No information	Some adaptations for sensitivity to the religious, socio-economic, and cultural backgrounds of the sample (no further information about the type of adaptation)
Hayati et al. (2018)	Peripheral treatment components – session structure	Designing shorter sessions of CBT (5-session CBT). According to the previous studies they found, shorter sessions to be more compatible with the Indonesian culture. On the contrary, the therapy process which is designed for up to 12 meetings, is often seen by Indonesians as too long, costly, complex, boring, and can reduce the trust in the therapist's abilities.
Subhas et al. (2021)	Core treatment components – modification	Integrating Islamic values into CBT by accommodating Islamic practice as aspects of the CBT technique, for example, practicing Muslim prayer as a form of meditation, referring to the Holy Quran and zikr (commemoration of Allah's name) as the way to enhance the spirituality, using Islamic based on gratitude, and inserting religion and spirituality when talking about the support system.
	Peripheral treatment components – materials and semantics Peripheral treatment components – cultural examples and themes	Translating the language to Bahasa Malaysia Referencing Prophet Muhammad's story as the example in the CBT because most Malaysian are Muslim. Changing in vivo situational exposure (from agoraphobic situations which they have avoided previously to a place close to his heart, the Mosque)
Wahyuningsih and Daengsari (2020)	No information	No detailed information was provided for the cultural adaptation they made.
Hayati and Tohari (2022)	No information	No detailed information was provided for the cultural adaptation they made.

Note: 2 studies modified core treatment components; 3 studies modified peripheral treatment components; 3 studies did not provide detailed information; 2 studies made cultural adaptations in multiple components

Table 3
The cultural adaptation process and framework used.

Authors	Cultural adaptation process and framework
Subhas et al. (2021)	Most of their adaptations in this study involved incorporating Islamic beliefs into CBT following: (1) a CBT manual that had been previously modified and adjusted following Malaysian society's norms and (2) general guidelines in "Religious-Cultural Psychotherapy in the Management of Anxiety Patients" by Razali and his colleagues
Drmic et al. (2017)	Emphasized the importance of a collaborative approach involving sharing decision-making between the investigators and key stakeholders at all phases of the implementation project. They used the traffic light model framework by the Centers for Disease Control and Prevention, Division of Reproductive Health, & ETR Associates. This framework was designed to guide intervention adaptations to accommodate the unique characteristics of local populations, including cultures.

translating the module and adjusting the more acceptable vocabulary for the patients (Drmic et al., 2017; Subhas et al., 2021). The second is modifying cultural examples and themes by embedding cultural examples, identities, values, beliefs, and practices into treatment (Drmic et al., 2017; Subhas et al., 2021). The third is modifying the session structure by changing the number of sessions (Hayati et al., 2018).

Based on this review, we conclude that cultural adaptations are insufficiently reported in several studies. Three of the seven studies provided incomplete information about the cultural adaptation type and component (Hayati and Tohari, 2022; Pityaratstian et al., 2007; Wahyuningsih and Daengsari, 2020). They did not elaborate on which specific CBT components were culturally adapted; they simply indicated that performing adaptation was made. Moreover, only two studies explicitly reported their framework and explained the adapting process in more detail (Drmic et al., 2017; Subhas et al., 2021).

Several reasons could be indicated why cultural adaptation is not sufficiently reported. First, no standard framework has been universally agreed upon and applied in research and practice (Rathod et al., 2018). In addition, the limited knowledge about cultural adaptation might be why several studies made cultural adaptation without a standardized framework. As a result, some researchers change or adapt without a transparent process. This finding aligns with a review of meta-analyses conducted by Rathod and colleagues in 2018 that revealed that from 12 articles found, most lacked methodological rigor and lacked consideration of underlying theory and cultural issues (Rathod et al., 2018). It becomes a challenge, and further research still needs to be developed. On the bright side, it may also be that some researchers have argued for the right balance between making new treatments for subgroups versus personalization (Bennett and Shafran, 2023).

The other possible reason is that the authors may not explicitly report on the process and content of cultural adaptations. On the one hand, standard documentation can lead to high-quality research with increased transparency on what, how, and why the cultural adaptation was made. Heim et al. (2021) proposed the Reporting Cultural Adaptation in Psychological Trials (RECAPT) criteria for reporting cultural adaptation in psychological trials. In RECAPT, each trial needs to document and report the following criteria: (1) set-up, (2) formative research methods, (3) intervention adaptation, and (4) measuring outcomes and implementation (Heim et al., 2021).

We could not continue to meta-analysis due to the scarcity of trials. This limitation indicates that research on CA-CBT for anxiety in Southeast Asian countries is still much needed. However, one study we found in this review (Bryant et al., 2011) reported that CA-CBT showed better improvement compared to the control group (treatment as usual

consisting of generic counseling). This aligns with previous research that showed culturally adapted intervention is effective and may even be more effective than unadapted intervention (Griner and Smith, 2006; Hall et al., 2016) and increases the acceptability of the treatment (Chowdhary et al., 2014). A case series study in India has also highlighted the need for culturally appropriate and acceptable forms of CBT to improve the outcome of CBT – one of them is by maximizing the non-specific factors in CBT (Samantaray et al., 2022).

There is still debate about the extent to which cultural adaptation should be made. The critical point is that keeping fidelity to the core components while incorporating local values and adaptations is crucial (Rathod et al., 2015). Moreover, Bennett and Shafran (2023) stated that evidence-based practice needs flexibility in intervention delivery, and not all modifications to therapy should be viewed as new treatments but instead regarded as meta-competence. Treatments always need adaptation to the context and preferences of an individual, also addressing individual differences within cultures or countries. Most experts suggested that adapting existing interventions into new contexts is preferable to designing new interventions (Bennett and Shafran, 2023). Such an approach would make implementing evidence-based treatments across different cultures much more feasible.

Several recommendations could be drawn from this research. First, further research focusing on cultural adaptation needs to be developed to get a comprehensive understanding. Second, trials must follow a standardized adaptation framework considering theoretical foundations. A clear guideline in reporting criteria is also vital to document and report all adaptation processes more transparently. Third, it is important to publish a separate article focusing on the cultural adaptation process rather than compiling the development in the same article with the trials. Explaining the cultural adaptation process in a separate paper enhances transparency, leaving more room for other important information in the trial paper.

This review has some strengths and limitations. First, the type of anxiety disorders varies, increasing the study's heterogeneity. This review also could not be continued to meta-analysis, so we could not determine the extent to which culturally adapted CBT is more effective than non-adapted CBT and which components have the most significant impact. Second, the quality of the studies, based on the risk of bias assessment, was quite heterogeneous, which may have affected the conclusions that can be drawn from the studies. Also, we did not consider grey literature in our review and only focused on the paper written in English and Indonesian, which might lead to selection bias.

Despite all the limitations, this paper has several strengths. First, we systematically searched databases and used a comprehensive method that followed Cochrane guidelines. We only included papers from peer-reviewed journals and assessed each paper's quality and bias using standardized tools. Second, although including the paper in Indonesian could result in bias, this could make the review sharper in picturing the condition in Indonesia.

In all, this study identified some examples of CBT adaptation in the context of anxiety disorders in Southeast Asia. Adaptations included both changes to the core components as well as to peripheral details such as language and case examples. Further studies will need to show whether such structural adaptations indeed lead to better treatment outcomes or whether we can invest in "meta-competence", training professionals to adapt the CBT framework to any individual rather than developing many local variants. Future studies could also extend the region to more than Southeast Asia and cover other psychiatric disorders (i.e., depressive disorders) to provide sufficient insight into the cultural adaptation of CBT.

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CRedit authorship contribution statement

Theo K. Bouman: Conceptualization, Formal analysis, Methodology, Supervision, Validation, Writing – review & editing. **Maaïke H. Nauta:** Conceptualization, Formal analysis, Methodology, Supervision, Validation, Writing – review & editing. **Ajeng Viska Icanervilia:** Conceptualization, Formal analysis, Investigation, Methodology, Validation, Writing – review & editing. **Ardian Praptomojati:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Project administration.

Declaration of Competing Interest

The authors declare no conflict of interest for the current study. MHN reports travel expenses, some subsistence, and speaker honoraria for lectures and clinical training workshops, mostly in the field of anxiety disorders in children and adolescents, and is a member of the national workgroup of the Dutch multi-disciplinary guideline for anxiety problems and disorders (non-remunerated).

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ajp.2023.103896](https://doi.org/10.1016/j.ajp.2023.103896).

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